ADA – Sensor Calibration Guide

Turning Raw Register Values into Real Temperatures

A Three-Point Field Calibration for the Temperature Sensors

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This sets out how I'm calibrating the six temperature sensors on the system – turning the raw number the PLC shows in a data register into a proper temperature reading in degrees Celsius, using a calibrated mercury thermometer and three water baths rather than any laboratory equipment. It also lays out where I'll record the results once the readings are taken.

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1 Purpose and Scope

The PLC already shows a number for every sensor – it just isn't, on its own, a trustworthy temperature. This guide covers how I'm turning that raw register value into an actual, accurate-as-practical reading in degrees Celsius, for all six temperature sensors on the system, using whatever's genuinely to hand rather than lab-grade kit.

The four level sensors go through similar hardware but calibrate against distance, not temperature – Section 9.2 covers that method and holds the print-ready tables for it, kept in this same document since the sampling protocol is identical either way.

A word on how tight this needs to be. This isn't an aircraft or a laboratory – it's a factory floor doing oil melting and retention. The tolerance doesn't need to be razor-thin, just good enough that the heater staging and demand logic downstream are working off numbers that mean something. Section 8 sets a working target rather than a precision-instrument spec.

2 Sensor and Signal-Path Inventory

2.1 What's Available

Hardware	Notes
2× PT100, built into the PLC	Ready to use
4-channel PT100 card	Not yet operational – haven't found a way to read it. If that changes, the plan below shifts accordingly.
2× analogue input, built into the PLC (AD2, AD3)	Ready to use
2× 4AD cards, 4 channels each (8 channels total)	Ready to use

2.2 The Current Plan

Sensors	Route into the PLC
2 temperature sensors	Built-in PT100
2 temperature sensors	Built-in analogue, AD3 and AD4
2 temperature sensors	4AD card (2 of its 4 channels)
4 level sensors	4AD card (all 4 channels of the second card)

Three different hardware paths feeding temperature into the PLC – PT100, built-in analogue, and card analogue – almost certainly means three different raw-to-real relationships, even for identical sensors. That's the main reason this can't just be one blanket conversion; each of the six temperature sensors gets its own calibration.

2.3 Wiring and Interface Connections

Temperature Sensor (SUP-WZPK Pt100 RTD) The 3-wire Pt100 RTD probe is adapted to a 2-wire configuration by bridging the two common return wires (Yellow), leaving a single Red signal wire and a single Yellow return wire.

- **Main CPU Inputs (Channels AD0 / AD1):** Connect the Red signal conductor to terminal `AD0` (or `AD1`) and the Yellow return conductor to `GND`.
- **Expansion Module (L02-4RTD):** Connect the Red signal conductor to terminal `TA`, the Yellow return conductor to terminal `TB`, and install a shorting link across terminals `TB` and `TC`.

Level Sensor (WSU180 CNRKA1WEXX Ultrasonic) The 2-wire 4–20 mA loop transmitter receives +24 V DC power on its positive supply terminal (+), with the loop current returning via the negative signal terminal (−).

- **Main CPU Inputs (Channels AD2 / AD3):** Connect the negative loop signal lead (−) to terminal `AD2` (or `AD3`), with system common referenced to the power supply 0 V DC (`GND`).
- **Expansion Module (L02-4AD):** Connect the negative loop signal lead (−) to terminal `V`, bridge terminals `I` and `V` with a shorting link to enable current sensing mode, and connect terminal `VI` to power supply 0 V DC.

3 Why Calibrate at All

Whatever number lands in the data register is a rough cousin of the actual temperature, not the actual temperature. Different sensor types, different cable runs, and the tolerances baked into each card all nudge that number away from what a thermometer would actually read. Calibrating just means working out, sensor by sensor, the relationship between "what the register says" and "what's really happening" – and then teaching the PLC that relationship.

4 Taming the Noise Before Calibrating

Raw sensor output jitters from scan to scan, so I'm not calibrating off a single reading – that would just bake the noise into the calibration. Instead:

- Timer T10 (preset K10, roughly a one-second tick) samples the raw value once a second.
- Five of those samples get summed and divided by five, giving a five-second average – the same trick already used for the process sensors elsewhere in the ADA ladder (Sensor Average 1 and 2).
- For each calibration point, I take **three** separate five-second averages, on both the thermometer and the PLC reading, rather than trusting any single average on its own.

5 The Three-Point Method

No lab, no calibration bath – just what's genuinely on the factory floor:

Point	Approximate role
Ice water	Cold anchor, near 0 °C
Normal water	Mid-range anchor, around ambient/tap temperature
Hot water (dispenser)	Hot anchor, well above ambient

A calibrated mercury thermometer is the reference throughout. It's a modest bit of kit, but it *is* a proper, calibrated reference, and that's what actually matters here – not how fancy the setup looks.

Per sensor: 3 points × 3 five-second-averaged readings × 2 instruments (thermometer and PLC) = 9 paired data points per sensor, across the full calibration.

6 The Maths – Fitting a Straight Line

The working assumption is that each sensor's response is linear across this range: a straight line

$$T_{\text{real}} = m \cdot x_{\text{raw}} + c$$

should describe the relationship between the raw register value x_{raw} and the true temperature T_{real} well enough. Three points is technically already one more than a straight line needs – two points define a line on their own – so the third point isn't just another anchor, it's a check on whether the straight-line assumption is actually holding, rather than something I'm just taking on faith.

With 9 paired readings per sensor rather than 3, a proper least-squares fit gives sturdier values for m and c than eyeballing a line through three dots, and the scatter around that line is a decent gut-check on linearity:

$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2}, \qquad c = \frac{\sum y - m\sum x}{n}$$

where x is the raw PLC reading, y is the thermometer reading, and $n = 9$ per sensor.

If the 9 points sit close to that fitted line, a straight-line calibration is a fair description of the sensor. If they don't, that's worth a second look before trusting the fit – possibly the sensor itself, possibly the readings.

7 Putting the Calibration Back Into the PLC

Once m and c are known for a sensor, they get applied the same way scaling is already done elsewhere in the ADA ladder: a multiply/divide pair for the gain m (since it's unlikely to be a clean integer), followed by an add or subtract for the offset c, writing the result into a proper, calibrated register kept separate from the raw one.

Rather than baking m and c into the ladder as fixed constants, I'm planning to store them as their own registers – the same tunable- without-a-program-change approach already used for the fault thresholds in the PLC Guide (D500–D514). That way a re-calibration later doesn't mean touching the ladder itself, just updating a couple of registers.

8 How Tight Does This Actually Need to Be

Worth saying plainly: this is a factory process, not a precision instrument. A working target of around $\pm 0.5\,^\circ$C feels sensible – tight enough that the heater staging and demand functions downstream are working off numbers that mean something, loose enough to be realistic given a mercury thermometer, an ice bucket, and a water dispenser.

This figure is a starting assumption, not a spec handed down from anywhere – worth checking against the actual scatter once real readings are in. If the fit comes in tighter than $0.5\,^\circ$C, good; if it doesn't, that's the number to revisit, not the method.

9 Calibration Data

Print-ready tables for all ten sensors – the six temperature sensors against the three-point method, and the four level sensors against a reference-point structure of their own. Fill in by hand, then transcribe into this file once done.

9.1 Temperature Sensors 1–6

Table 1: Sensor 1 (PT100, built-in) – calibration readings.

Thermometer (°C)	PLC (raw)	Predicted (°C)	Residual
1	24	1.56	−0.56
35	358	33.33	+1.67
48	524	49.12	−1.12

Fitted result: $m = 0.095124$ $c = -0.7275$ $R^2 = 0.996$ max residual 1.67°C

Table 2: Sensor 2 (PT100, built-in) – calibration readings.

Thermometer (°C)	PLC (raw)	Predicted (°C)	Residual
2	39	1.96	+0.04
1	30	1.03	−0.03
32	345	33.38	−1.38
34	344	33.28	+0.72
34	345	33.38	+0.62
48	487	47.97	+0.03

Fitted result: $m = 0.102705$ $c = -2.0502$ $R^2 = 0.999$ max residual 1.38°C

Table 3: Sensor 3 (built-in analogue, AD3) – calibration readings.

Thermometer (°C)	PLC (raw)	Predicted (°C)	Residual
1	15	0.59	+0.41
30	323	30.93	−0.93
55	562	54.48	+0.52

Fitted result: $m = 0.098505$ $c = -0.8849$ $R^2 = 0.999$ max residual 0.93°C

Table 4: Sensor 4 (built-in analogue, AD4) – calibration readings.

Thermometer (°C)	PLC (raw)	Predicted (°C)	Residual
1	27	1.00	0.00
30	317	30.00	0.00
50	517	50.00	0.00

Fitted result: $m = 0.1$ $c = -1.7$ $R^2 = 1.000$ max residual 0.00°C – the cleanest sensor of the six; $T = 0.1x - 1.7$ falls exactly on all three points.

Table 5: Sensor 5 (4AD card, channel 1) – calibration readings.

Thermometer (°C)	PLC (raw)	Predicted (°C)	Residual
1	17	1.06	−0.06
30	304	29.85	+0.15
46	466	46.10	−0.10

Fitted result: $m = 0.100318$ $c = -0.6500$ $R^2 = 1.000$ max residual 0.15°C – comfortably inside the ±0.5°C target.

Table 6: Sensor 6 (4AD card, channel 2) – calibration readings.

Thermometer (°C)	PLC (raw)	Predicted (°C)	Residual
0	13	−2.02	+2.02
24	300	27.39	−3.39
72	722	70.63	+1.37

Fitted result: $m = 0.102462$ $c = -3.3493$ $R^2 = 0.994$ max residual 3.39°C

Corrected – the 72°C reading was 722, not 122. A typo, as suspected: 722 sits close to the ∼720–750 the other sensors' gain would predict. With that fixed, Sensor 6 fits its own line reasonably well ($R^2 = 0.994$). The remaining outlier is the **24°C point** (raw 300), sitting 3.4°C off Sensor 6's own fitted line – worth a repeat reading there specifically, whenever there's a spare moment, rather than treating the sensor as suspect.

One shared formula, not six separate ones. Sensors 4 and 5 are taken as the trusted anchors – pooling their six readings together (rather than picking either sensor's own line outright, which would be perfect for that one and worse for the other) gives:

$$m = 0.09996 \approx 0.1, \qquad c = -1.1223$$

The gain landing almost exactly on 0.1 is a useful accident: it lines up with the $\times 10$ fixed-point convention already used throughout the ADA ladder (D150 stores $T_{\text{target}} \times 10$, and so on). Read the other way round, the raw register already reads roughly ten times the real temperature – which means, in practice, this calibration is mostly about finding the *offset*, not the gain.

Rounding c to -11 on a $\times 10$ scale gives a formula simple enough to implement as a single subtraction:

$$T \times 10 = x_{\text{raw}} - 11$$

One `SUB` instruction, no multiply or divide at all – about as cheap as PLC maths gets, and well within the accuracy this method can support in the first place.

Table 7: Error under the shared formula ($T = 0.1x - 1.13$), per sensor.

Sensor	Error at each calibration point
1	$+0.3$, -0.3, $+3.3$ °C
2	$+0.8$, $+0.9$, $+1.4$, -0.7, -0.6, -0.4 °C
3	-0.6, $+1.2$, $+0.1$ °C
4	$+0.6$ °C at all three points (consistent, not zero – the shared formula is a compromise between 4 and 5, not a perfect fit to either)
5	-0.4 to -0.7 °C across its three points
6	$+0.2$ °C at 0°C, $+4.9$ °C at 24°C, -0.9 °C at 72°C

Most sensors sit within a couple of degrees of the shared formula, which for a hand-held thermometer against a wobbling probe with no lab kit is a reasonable place to land. Two points are worth a second look, not because they "failed" anything, but because they stand out from the rest of the spread: Sensor 1's third reading (+3.3°C) and Sensor 6's second reading (+4.9°C) are both noticeably larger than every other error in the table. Worth a repeat measurement at those two points specifically when there's a spare moment, rather than treating the whole sensor as suspect.

9.2 Level Sensors 7–10

Method, as run. Each sensor was fixed facing sideways at a known reflective target – a control panel cover and a chair, positioned to give three distinct, adjustable distances between the sensor and that reflective body. Sampling follows the same protocol as the temperature sensors: a 1-second sample tick, five samples averaged into a 5-second reading, three such 5-second averages taken at each distance. The distance recorded is sensor-to-body, *not* the quantity the process actually cares about – Section 9.3 covers turning that into a tank level.

Sensors 7 and 8 read through the built-in analogue channels (AD2/AD3), so their PLC value is a direct 5-second average, an integer. Sensors 9 and 10 read through the 4AD card, where the ladder already divides the raw value by 10 before it lands in the register, so their PLC value is a float roughly a tenth the size of sensors 7/8's for a comparable distance.

Table 8: Sensor 7 (level, built-in analogue, direct average) – calibration readings.

Distance (mm)	PLC (raw)	Predicted (mm)	Residual
53	3454	50.35	+2.65
53	3467	47.59	+5.41
53	3453	50.56	+2.44
96	3161	112.43	−16.43
96	3169	110.73	−14.73
96	3171	110.31	−14.31
144	3059	134.04	+9.96
144	3101	125.14	+18.86
144	3041	137.85	+6.15

Fitted result: $m = -0.211885$ $c = 782.1949$ $R^2 = 0.901$ max residual 18.86 mm

Sensor 7's fit is the loosest of the four – every reading at the **96 mm** point lands 14–16 mm below the line, on its own, in the same direction, which looks less like scatter and more like something systematic at that particular distance (target angle, a stray reflection off the panel edge, or genuine non-linearity in this sensor over this range). Worth a repeat set at 96 mm specifically before trusting this fit for anything tight.

Table 9: Sensor 8 (level, built-in analogue, direct average) – calibration readings.

Distance (mm)	PLC (raw)	Predicted (mm)	Residual
56	3411	51.66	+4.34
56	3428	48.38	+7.62
56	3408	52.24	+3.76
108	3062	118.96	−10.96
108	3030	125.13	−17.13
108	3058	119.73	−11.73
177	2809	167.75	+9.25
177	2790	171.41	+5.59
177	2809	167.75	+9.25

Fitted result: $m = -0.192840$ $c = 709.4354$ $R^2 = 0.962$ max residual 17.13 mm

Sensor 8 shows the same pattern as Sensor 7 – the **middle** distance (108 mm) is where the fit sags the most, again all three reps in the same direction. The two built-in-analogue sensors bracketing their worst error at the same relative point (mid-range) rather than at the near or far end suggests it's worth checking the fixture geometry at that middle distance specifically, rather than treating it as two unrelated coincidences.

Table 10: Sensor 9 (level, 4AD card, ÷10 scaling) – calibration readings.

Distance (mm)	PLC (raw)	Predicted (mm)	Residual
58	2787.56	55.08	+2.92
58	2788.04	54.98	+3.02
58	2790.26	54.49	+3.51
103	2548.30	107.94	−4.94
103	2547.26	108.17	−5.17
103	2538.12	110.19	−7.19
178	2253.00	173.19	+4.81
178	2232.78	177.66	+0.34
178	2243.46	175.30	+2.70

Fitted result: $m = -0.220945$ $c = 670.9779$ $R^2 = 0.993$ max residual 7.19 mm

Table 11: Sensor 10 (level, 4AD card, ÷10 scaling) – calibration readings.

Distance (mm)	PLC (raw)	Predicted (mm)	Residual
57	2787.60	57.02	−0.02
57	2796.38	54.97	+2.03
57	2790.40	56.37	+0.63
131	2459.28	133.47	−2.47
131	2458.76	133.59	−2.59
131	2459.06	133.52	−2.52
177	2279.22	175.39	+1.61
177	2279.06	175.43	+1.57
177	2279.88	175.24	+1.76

Fitted result: $m = -0.232846$ $c = 706.1015$ $R^2 = 0.999$ max residual 2.59 mm – the cleanest of the four level sensors, on a par with the best temperature sensors.

The 4AD-card pair (9, 10) clearly out-performs the built-in pair (7, 8). R^2 runs 0.993–0.999 for Sensors 9/10 versus 0.901–0.962 for Sensors 7/8, and the worst residual on the card pair (7.2 mm) is smaller than the *best* residual on the built-in pair (17.1 mm). Whether that's the channel type itself, the ÷10 pre-scaling smoothing things out, or just where they happened to sit relative to the reflective target isn't clear from three points – but if Sensors 7/8's readings hold up under a repeat set, it's worth treating them as the less trustworthy pair pending a wider tolerance or a re-check of their fixture.

9.3 From Sensor-to-Body Distance to Tank Level

None of the four fits above give a tank level directly – they give the distance from the sensor to whatever it's facing, which in the field setup was a control panel cover and a chair, not the liquid surface. With the sensor mounted facing down into the tank and h the fixed distance from the sensor to the bottom of the tank (or to empty/zero level, however that's defined for the tank in question), the liquid level is what's *left over* once the sensed distance is subtracted:

$$L = h - (m \cdot x_{\text{raw}} + c)$$

where L is the liquid level (mm, measured up from the tank's zero point), h is that fixed sensor-to-zero-point distance for the tank the sensor is mounted on, and m, c are the fitted values for that specific sensor from the tables above.

h isn't a calibration quantity – it's a one-off measurement of the physical mounting (sensor height above the tank's empty-level point), taken once per tank rather than fitted from data, so it isn't included in the tables above. Until that measurement is taken and dropped in, m and c alone give a usable distance reading; the h subtraction is the last step that turns it into a level.

10 The Conversion Equation, Plainly

All the maths in Section 7 boils down to a short recipe. Once m and c are known for a sensor, getting from the raw register value to a real temperature is just:

1. **Take the raw reading** straight from its register.
2. **Multiply it** by the gain, m – scaled up first (by 100 or 1000) if m isn't a clean whole number, since the PLC's integer maths doesn't handle decimals directly.
3. **Divide** by whatever that scaling factor was, to bring the result back down to the right size.
4. **Add (or subtract) the offset**, c – add if c came out positive, subtract if it came out negative.
5. **That's the calibrated temperature**, ready to store in its own register, separate from the raw one.

In short: **multiply by the gain, divide back down, then add the offset.** Three PLC instructions, same shape as the scaling already done elsewhere in the ADA ladder – nothing new to invent, just new numbers to drop in once the tables above are filled in.

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